Modified Cuckoo Search Algorithm to Solve Economic Power Dispatch Optimization Problems

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***Abstract*—A modified cuckoo search (CS) algorithm is proposed to solve economic dispatch (ED) problems that have non-convex, non-continuous or non-linear solution spaces considering valve-point effects, prohibited operating zones, transmission losses and ramp rate limits. Comparing with the traditional cuckoo search algorithm, we propose a self-adaptive step size and some neighbor-study strategies to enhance search performance. Moreover, an improved lambda iteration strategy is used to generate new solutions. To show the superiority of the proposed algorithm over several classic algorithms, four systems with different benchmarks are tested. The results show its efficiency to solve economic dispatch problems, especially for large-scale systems.**

***Index Terms*—Cuckoo search (CS), economic dispatch (ED), prohibited operating zones, ramp rate limits, valve-point effects.**

I. Introduction

ECONOMIC dispatch (ED) is one of the most fundamental optimization problems in electric power systems with the objective to minimize the total cost for power generation. It aims at economically allocating the load demand among the generators while satisfying several equality and inequality constraints in the systems.

As a classical optimization problem, ED with smooth cost functions has been solved by numerous traditional programming methods such as gradient methods [1], lambda iteration method [2], quadratic programming [3], linear programming [4], dynamic programming [5] and Lagrangian method [6]. In recent years, several ED problems with some complex and non-smooth functions are proposed by considering transmission network losses and plant constraints such as valve-point effects, multiple fuel options, generation ramp rates, and prohibited operation zones. Most of the traditional techniques are not capable of efficiently solving such problems that have non-convex, non-continuous or non-linear solution spaces. Over the past two decades, evolutionary computation developed rapidly [7]−[14]. Many modern meta-heuristic algorithms and their variants were successfully used to solve such problems. According to their characteristics, they can be divided into three types: evolutionary algorithms [15]−[19], simulated ecosystem algorithms [20]−[24], and swarm intelligence algorithms [25]−[31].

Cuckoo search (CS) algorithm is a nature-inspired swarm intelligence technique based on the brood parasitism of some cuckoo species, as introduced by Yang and Deb in 2009 [32]. Due to its simple concept and easy implementation, CS has been successfully applied to tackle uni-modal and multi-modal numerical optimization and engineering problems [33]−[41]. Many researchers have applied CS to solve ED problems in power systems [9], [42]−[48]. Several studies show that CS can always find the optimal results, but it may not guarantee a fast convergence because its searching process relies entirely on a random walk [9], [36], [47]. Meanwhile, a small or regular step may cause it to be trapped in a local optimal solution [48]. To overcome this deficiency, a modified cuckoo search algorithm (MCSA) is proposed in this paper, where a neighbor study strategy is designed and a self-adaptive parameter selection strategy is formulated. Compared with the existing studies, we have made three contributions:

1) A new self-adaptive step size strategy is proposed such that the step decreases in different speeds as iterations proceed. In the beginning, it maintains a high value in order to enhance the exploration ability. Then, it declines rapidly to its minimum value in order to make MCSA converge steadily to a refined solution.

2) A neighbor study strategy is adopted. When the best solution is no longer updated after a number of iterations, each

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solution can exchange the information with others randomly.

3) A new lambda iteration method is designed to generate feasible solutions at the initial stage. In MCSA, all solutions must be feasible. A relaxation method is thus designed to handle the equality constraint that may lead to infeasible solutions.

The rest of paper is organized as follows: Section II describes the ED problem. The standard CS and MCSA are introduced in Section III. Section IV implements MCSA to solve the ED problem. Section V is dedicated to numerical simulations and results. Conclusions and future work are given in Section VI.

II. Problem Formulation

The problem discussed in this paper is the same as those in literature [15]−[20], [26], [43]. The objective of ED problem is to minimize the fuel cost of generators in electric power systems for a given load demand subject to various constraints.

A. Objective Function

The fuel function without valve-point loading of generators is given below.

$$\min F_t = \sum_{i=1}^{D} F_i(P_i) = \sum_{i=1}^{D} a_i + b_i P_i + c_i P_i^2 \tag{1}$$

where D is the total number of generators. $F_i(P_i)$ is the fuel cost of the ith generator with unit \$/h. P_i is the power in megawatt (MW) generated by the ith generator, and a_i, b_i and c_i are respectively the cost coefficients of the ith generator.

Practical large-size generators usually have multi-valve steam turbines. When each steam valve is on or off, it may produce a ripple. Usually, a sinusoidal term is added in (1) with consideration of valve-point effects (VPE) [2], thus leading to

$$\min F_t = \sum_{i=1}^{D} a_i + b_i P_i + c_i P_i^2 + \left| e_i \sin\left(f_i\left(P_i^{\min} - P_i\right)\right)\right| \tag{2}$$

where e_i and f_i are constants of the valve-point effects of generators.

B. Equality Constraint

In order to balance the power, the total generated power should meet the power demand and transmission loss (TL)

$$\sum_{i=1}^{D} P_i = P_T + P_L \tag{3}$$

where P_T is the total power demand in MW, and P_L represents the transmission losses in MW which can be computed by using B-coefficients [2] and is given by

$$P_L = \sum_{i=1}^{D}\sum_{j=1}^{D} P_i B_{ij} P_j + \sum_{i=1}^{D} B_{0i} P_i + B_{00} \tag{4}$$

where B_{ij}, B_{0i} and B_{00} are the loss coefficients which are constant under normal operational conditions.

C. Inequality Constraint

The output power of each generator has a lower limit and an upper one

$$P_i^{\min} \le P_i \le P_i^{\max}, \quad i = 1, 2, \ldots, D \tag{5}$$

where $P_i^{\min}$ and $P_i^{\max}$ are the minimum and maximum power, respectively, in MW generated by the ith generator.

D. Prohibited Operating Zones

Each generator may have certain prohibited operating zones (POZ) caused by opening or closing its steam valve. The feasible operating zones of generator i can be described as follows:

$$P_i \in \begin{cases} P_i^{\min} \le P_i \le P_{i,1}^{l} \\ P_{i,j-1}^{u} \le P_i \le P_{i,j}^{l} \\ P_{i,n_j}^{u} \le P_i \le P_i^{\max} \end{cases}, \quad j = 2, 3, \ldots, n_j \tag{6}$$

where n_j is the number of POZ of the ith generator, and $P_{i,j}^{l}$ and $P_{i,j}^{u}$ are the lower and upper bounds of power in the jth POZ by the ith generator, respectively.

E. Ramp Rate Limits

Practically, all generators should satisfy the physical limitation of starting up and shutting down by using ramp rate limits (RRL). The increase and reduction of power generation in each generator are limited by

$$P_i - P_i^0 \le U_i \tag{7}$$

$$P_i^0 - P_i \le L_i \tag{8}$$

where P_i^0 is the previous output power. U_i and L_i are the up-ramp limit and the down-ramp limit of the ith generator, respectively.

Combining (7) and (8) with (5) results in the change of the generation limits to

$$\underline{P}_i \le P_i \le \overline{P}_i, \quad i = 1, 2, \ldots, D \tag{9}$$

where

$$\underline{P}_i = \max(P_i^{\min}, P_i^0 - L_i), \quad \overline{P}_i = \min(P_i^{\max}, P_i^0 + U_i).$$

III. Modified Cuckoo Search Algorithm

A. Standard Cuckoo Search (CS) Algorithm

CS is a population-based swarm intelligence algorithm inspired by the interesting breeding behavior of cuckoo [32]. It is enhanced by the so-called Lévy flight, rather than by simple isotropic random walks. There are mainly three rules during its searching process [49]. Its main steps are given below:

1) Initialization

Suppose that there are N host nests with D dimensions. A population of these host nests could be denoted by a vector as $X = [X_1, X_2, \ldots, X_N]^T$ where $X_k = [X_k^1, X_k^2, \ldots, X_k^D]^T$. Each solution is randomly generated within the boundary range of the parameters and given by

$$X_k^i = P_i^{\min} + r_1 \times \left(P_i^{\max} - P_i^{\min}\right) \tag{10}$$

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where $k = 1, 2, \ldots$, and N represents the index of a nest in the population and $i = 1, 2, \ldots$, and D represents the ith generator. r_1 is a uniformly distributed random number between 0 and 1.

2) New solution generation via Lévy flight

After initialization, CS uses a Lévy flight random walk to search a new solution, denoted by $X_{k(\text{new})}$. In order to calculate the optimal step length for the Lévy flight, one of the most effective ways is to use the Mantegna algorithm with a symmetric Lévy stable distribution [49], [50]. The new solution for each nest can be formulated as follows:

$$X^i_{k(\text{new})} = X^i_k + \alpha \times r_2 \times \Delta X^i_{k(\text{new})} \tag{11}$$

where α is the step size, and is usually set to be 0.01. r_2 is a random number that satisfies the standard normal distribution. $\Delta X^i_{k(\text{new})}$ is calculated as follows:

$$\Delta X^i_{k(\text{new})} = \frac{u}{|v|^{\frac{1}{\beta}}} \times \frac{\sigma_u(\beta)}{\sigma_v(\beta)} \times \left(X^i_k - X^i_{\text{gbest}}\right) \tag{12}$$

$$\sigma_u(\beta) = \left\{ \frac{\Gamma(1+\beta) \times \sin\left(\pi \times \frac{\beta}{2}\right)}{\Gamma\left[\frac{1+\beta}{2}\right] \times \beta \times 2^{\frac{\beta-1}{2}}} \right\}^{\frac{1}{\beta}}, \quad \sigma_v(\beta) = 1 \tag{13}$$

where u and v are two standard normally distributed stochastic variables with standard deviations σ_u and σ_v, respectively. β is the distribution factor satisfying $0 < \beta \leq 2$ and it is set equal to 1.5 in this work, Γ is a gamma distribution function, and X_{gbest} is the best nest in the swarm.

3) A new solution generated via a crossover operation

The discovery of an alien egg in a nest of a host bird has a probability denoted by P_α. A new solution $X_{k'}$ is computed as

$$X^i_{k'} = \begin{cases} X^i_k + r_3 \times \left(X^i_{c1} - X^i_{c2}\right), & \text{if } r_4 > P_a \\ X^i_k, & \text{otherwise} \end{cases} \quad i = 1, 2, \ldots, D \tag{14}$$

where r_3 and r_4 are random numbers that are uniformly distributed between 0 and 1. X_{c1} and X_{c2} are randomly chosen from the swarm.

The pseudo code of CS is given as Algorithm 1.

Algorithm 1 The Standard CS Algorithm

1. Initialize a population of host nests
2. Get the global best solution
3. **While** (stop criterion = false) **Do**
4. Update the solution via Lévy flight, compare with the last generation, and keep the current individual as the best nest
5. Replace a fraction of eggs to generate a new solution via a crossover operation
6. Keep the current individual solution as the best one by comparing the solutions before and after a crossover operation
7. Get the global solution
8. **End while**

B. Self-Adaptive Step Size

α is a very important parameter for CS to find globally best solutions. CS uses a fixed value, for example $\alpha = 0.01$. In fact, larger α can enhance its exploration ability and avoid its falling into a local optimum in its beginning stage. As iterations proceed, smaller α makes CS converge steadily to a better solution. Inspired by [30], α is calculated as

$$\alpha(t) = b - a \times \frac{\exp\left(\frac{10 \times (t-1)}{t^M - 1}\right) - 1}{\exp(10) - 1}, \quad t = 1, 2, \ldots, t^M \tag{15}$$

where t^M represents the maximum number of iterations, t means iteration t, and a and b are two positive constants satisfying $a < b < 1$. Fig. 1 presents an example of $\alpha(t)$. Its value is ranging from 0.4 to 0.01, when $a = 0.39$ and $b = 0.4$.

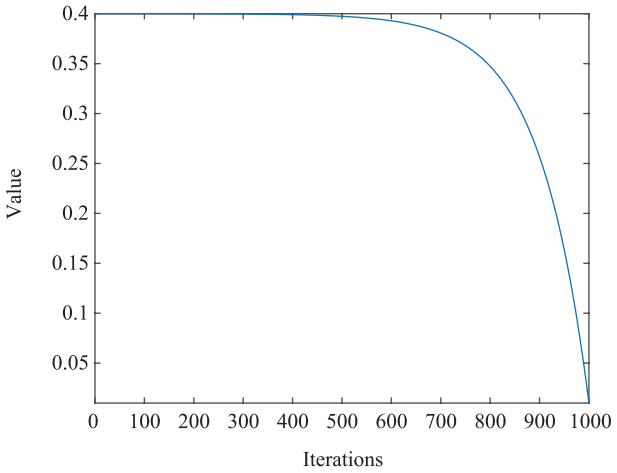


Fig. 1. Value of $\alpha(t)$ from 0.4 to 0.01, when $a = 0.39$ and $b = 0.4$.

C. Neighbor Study Strategy

In a standard CS, new solutions are generated via (14). It is obvious that solutions may easily be trapped in a local optimum if the search environment is complex with local optima. A solution may have the same values as the global optimum in some dimensions, but it has a high fitness value for a minimization problem because of the poor solutions in the other dimensions [30]. In order to make a better use of the beneficial information, a neighbor study strategy is proposed to improve CS as inspired by the work [30], [31], [51]. $X_{k'}$ is calculated as follows:

$$X^i_{k'} = (1 - K) \times X^i_k + K \times \gamma^i_k, \quad i = 1, 2, \ldots, D \tag{16}$$

where K is an updated coefficient based on the probability of discovering an alien egg in its nest and is calculated as

$$K = \begin{cases} 1, & \text{if } r_5 > P_\alpha \\ 0, & \text{otherwise} \end{cases} \tag{17}$$

where r_5 is a uniformly distributed random number between 0 and 1; and P_α is the probability value of discovering an alien egg and $P_\alpha = 0.25$. γ^i_k is the exemplar for X^i_k, which is obtained from the other two solutions' competition in the ith dimension.

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Every solution in the nest is able to learn from other solutions' best experiences in different dimensions. Thus, the ability of exploration is enhanced by such information sharing mechanism. Algorithm 2 describes the method of generating an exemplar $\Upsilon = [\gamma_1, \gamma_2, \ldots, \gamma_N]^T$. In order to ensure that a solution learns from good exemplars and minimize the time wasted on poor directions, Υ is rebuilt if the best global solution is not changed in the next 3 consecutive iterations.

Algorithm 2 Generate Exemplar Dimensions for X_k

```
Input X_k
For i = 1 to D
  If K = 1
    Sol_1 and Sol_2 are two solutions randomly selected
    from host nests
      If fit(Sol_1) < fit(Sol_2)
        γ_k^i = sol_1^i
      Else
        γ_k^i = sol_2^i
      End if
  Else
    γ_k^i = X_k^i
  End if
End for
Return γ_k
```

IV. MCSA to Solve ED Problems

In this section, MCSA is applied to solve the ED problem. First, a new solution-generated method is introduced. Then, inequality and equality constraints' handling techniques are proposed. Finally, the main steps of MCSA are described.

A. Modified Lambda Iteration Method

In the standard CS, a solution generated via (10) is usually infeasible and difficult to repair especially in large-scale systems. Thus it is necessary to propose a fast and reasonable method to solve this infeasibility problem. The classical lambda iteration was introduced in literature and applied in many software packages [2]. Although it is widely used by power utilities for ED, improper selection of the initial value may cause slow convergence and sometimes leads to divergence [52]−[54]. In this section we propose a method to generate a new solution effectively. Although the new solution is sometimes infeasible, it is easier to repair compared to the traditional methods especially in a system with many generators.

At the ith generator,

$$F_i(P_i) = a_i + b_i P_i + c_i P_i^2 \qquad P_{i,\min} \le P_i \le P_{i,\max}, \quad i = 1, 2, \ldots, D \tag{18}$$

$$\frac{dF_i}{dP_i} = \lambda_i = b_i + 2c_i P_i. \tag{19}$$

Then, we calculate $\lambda_{i,\min}$ and $\lambda_{i,\max}$, $i = 1, 2, \ldots, D$ as follows:

$$\lambda_{i,\min} = b_i + 2c_i P_i, \quad \text{at } P_i = P_{i,\min} \tag{20}$$

$$\lambda_{i,\max} = b_i + 2c_i P_i, \quad \text{at } P_i = P_{i,\max}. \tag{21}$$

At last, we calculate λ_{mean} and λ_{var} via

$$\lambda_{\text{mean}} = \frac{1}{2D} \sum_{i=1}^{D} (\lambda_{i,\min} + \lambda_{i,\max}) \tag{22}$$

$$\lambda_{\text{var}} = \frac{1}{2D} \sum_{i=1}^{D} \left((\lambda_{i,\min} - \lambda_{\text{mean}})^2 + (\lambda_{i,\max} - \lambda_{\text{mean}})^2 \right). \tag{23}$$

The method to generate a new solution is given in Algorithm 3.

Algorithm 3 Generate a New Solution

```
Calculate λ = λ_mean + randn × sqrt(λ_var)
For i = 1 to D
  If λ < λ_{i,min}
    temp = λ_{i,min} + 0.5 × rand × (λ_{i,max} − λ_{i,min})
  Else if λ > λ_{i,max}
    temp = λ_{i,max} + 0.5 × rand × (λ_{i,max} − λ_{i,min})
  Else
    temp = λ
  End if
  X_k^i = (temp − b_i)/(2c_i)
End for
Return X_k
```

In Algorithm 3, sqrt represents the square root, randn represents a normally distributed pseudorandom number, and rand represents a uniformly distributed pseudorandom number.

B. Constraint Handling Mechanism

1) Inequality Constraints

The global solutions should satisfy inequality constraint (9). During the searching process, if there are some solutions that are not in the scope of the feasible solution region, MCSA may stop at the region boundary.

For constraint (6), when a unit operation is in one of its POZ, a repairing strategy is activated [43]. P_i violating its prohibited zone is adjusted via

$$P_i^{\text{new}} = \begin{cases} P_{i,j}^l, & \text{if } P_i \le P_{i,j}^m \\ P_{i,j}^u, & \text{if } P_i > P_{i,j}^m \end{cases} \qquad i = 1, 2, \ldots, D, \quad j = 2, 3, \ldots, n_j \tag{24}$$

where $P_{i,j}^m$ is the midpoint of each prohibited zone and computed as $P_{i,j}^m = (P_{i,j}^l + P_{i,j}^u)/2$.

2) Equality Constraints

Although a solution satisfies all inequality constraints, it may be infeasible if it does not satisfy the power balance constraint (3). When it happens, the simplest approach for handling such infeasible solutions is to use the penalty function. However, it is well known that defining a proper penalty coefficient is difficult. Hence, a slack approach [42], [43] is used in this paper.

Assume that a slack generator s $(1 \le s \le D)$ is arbitrarily selected and the power outputs of the remaining D-1 genera-

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tors are known. The power output of the slack generator s is calculated as

$$P_s = P_T + P_L - \sum_{i=1,i\neq s}^{D} P_i. \tag{25}$$

The transmission loss P_L in (3) can be expressed by considering P_s as an unknown variable

$$P_L = B_{ss}P_s^2 + \left(2\sum_{i=1,i\neq s}^{D} B_{si}P_i + B_{0s}\right) \times P_s + \sum_{i=1,i\neq s}^{D}\sum_{j=1,j\neq s}^{D} P_i B_{ij} P_j + \sum_{i=1,i\neq s}^{D} B_{0i}P_i + B_{00}. \tag{26}$$

By substituting (26) into (25), we have

$$A \times P_s^2 + B \times P_s + C = 0 \tag{27}$$

where coefficients A, B and C are given as

$$A = B_{ss} \tag{28}$$

$$B = 2\sum_{i=1,i\neq s}^{D} B_{si}P_i + B_{0s} - 1 \tag{29}$$

$$C = \sum_{i=1,i\neq s}^{D}\sum_{j=1,j\neq s}^{D} P_i B_{ij} P_j + \sum_{i=1,i\neq s}^{D} B_{0i}P_i + B_{00} + P_T - \sum_{i=1,i\neq s}^{D} P_i. \tag{30}$$

The power output of can be calculated by solving (27). The pseudo code is shown in Algorithm 4.

Algorithm 4 Calculate P_s Considering TL

1. X_k is generated by Algorithm 3 and adjusted by (24); it satisfies constraints (6) and (9), but does not satisfy constraint (3); let S denote the slack unit number; calculate A, B, C and $\Delta = B^2 - 4AC$
2. **If** $\Delta < 0$
3. **Return** 0
4. **Else**
5. $x_{1,2} = (-B \pm \mathrm{sqrt}(\Delta))/2$
6. **If** x_1 satisfies constraints (6) and (9)
7. $X_k^s = x_1$
8. **Return** X_k
9. **Else if** x_2 satisfies constraints (6) and (9)
10. $X_k^s = x_1$
11. **Return** X_k
12. **End if**
13. **End if**

If the return value is 0, Algorithm 4 is re-executed until solution X_k becomes feasible. In addition, although some systems consider no transmission loss, the slack method is valid and P_s can be solved as

$$P_s = P_T - \sum_{i=1,i\neq s}^{D} P_i. \tag{31}$$

3) Application of MCSA for ED

In MCSA, each nest represents a solution and a population of nests are used for finding the best solution of the problem. The main steps of MCSA are illustrated in Fig. 2.

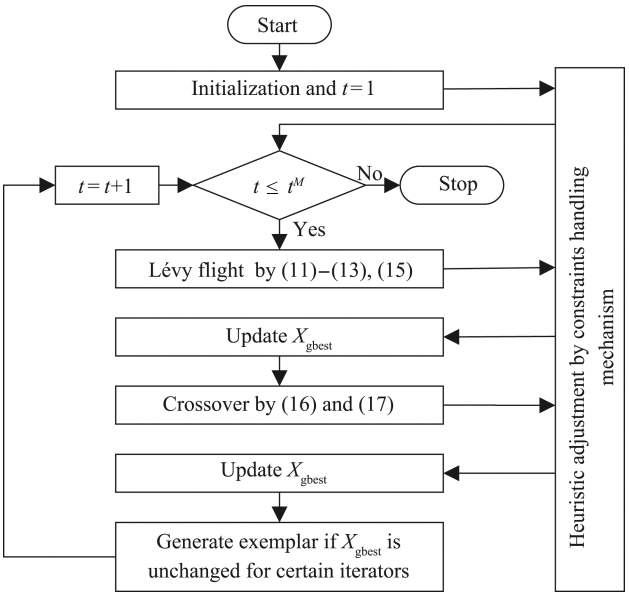


Fig. 2. Flow chart of MCSA.

V. Case Study

In order to demonstrate the efficiency and robustness of MCSA for solving the ED problem, some cases are conducted on 6, 13, 20, and 40-unit systems, and the results are compared with several state-of-the-art ED algorithms in the literature. All case studies are implemented in MATLAB R2016a, on a personal computer with Intel i5 2.3 GHz processor, 4 GB of RAM and Windows 10 Professional. Due to the stochastic nature of an evolutionary algorithm in each case, 50 independent trials are conducted to calculate the best, mean, and worst fuel costs, and its standard deviation for each test system.

A. Parameter Selection

The performance of MCSA is sensitive to parameter settings. In this work, the parameters of MCSA described in Table I are selected based on a rigorous empirical study for each case. In this paper, the value of the self-adaptive step size α is ranging from 0.4 to 0.01 for all tested systems with $b = 0.4$ and $a = 0.39$ in (15). The parameters of generator count numbers (D), demanded load (P_T), solution count in population (N), P_α, and maximum iteration count (t^M) for all tested systems are described in Table I. In addition, the characteristics of each system such as prohibited operating zones (POZ), transmission losses (TL), valve-point effects (VPE), and ramp rate limits (RRL) are also shown in the table.

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TABLE I
CHARACTERISTICS FOR ALL CASES

Cases	D	P_T (MW)	N	P_α	POZ	TL	VPE	RRL	t^M
Case 1	6-unit	1263	25	0.25	Yes	Yes	No	Yes	200
Case 2	13-unit	2520	25	0.25	No	No	Yes	No	1000
Case 3	13-unit	2520	25	0.25	No	Yes	Yes	No	1000
Case 4	20-unit	2500	40	0.25	No	Yes	No	No	1000
Case 5	40-unit	10 500	40	0.65	No	No	Yes	No	2000

B. System 1 (Case 1: 6-unit)

The first tested system is a 6-unit system which has a demand of 1263 MW with POZ, TL and RRL. Its input data are taken from [55], [56]. The objective function for this system is smooth and no convexity is given by the prohibited operating zones and ramp rate limits. The best generation values, transmission losses and optimal cost obtained by MCSA are presented in Table II. Note that all system constraints, such as POZ and RRL are satisfied. The total generation cost and the corresponding transmission loss are 15 449.8995 $/h and 12.9582 MW, respectively.

TABLE II
OPTIMAL GENERATIONS AND COSTS OBTAINED BY MCSA FOR CASE 1

Unit	$\underline{P}_i$ (MW)	$\overline{P}^i$ (MW)	POZ	Generation (MW)
1	320	500	[210,240]; [350,380]	447.5038
2	80	200	[90,110]; [140,160]	173.3182
3	100	265	[150,170]; [210,240]	263.4628
4	60	150	[80,90]; [110,120]	139.0653
5	100	200	[90,110]; [140,150]	165.4734
6	50	120	[75,85]; [100,105]	87.1347
Cost ($/h)			15 449.8995	
Transmission loss (MW)			12.9582	

Table III shows the obtained best cost, mean cost, worst cost, standard deviation and time for this test system after 50 trial runs. To show the differences among the results, we use four digits after the decimal point. These results are compared with other algorithms that have been reported recently such as modified artificial bee colony (MABC) [57], backtracking search algorithm (BSA) [58], differential harmony search (DHS) [59], new particle swarm optimization with local random search (NPSO-LRS) [60], particle swarm optimization (PSO) [55], multiple tabu search (MTS) [61], chaotic bat algorithm (CBA) [62], and CS [63]. TOL is the difference of power balance calculated by

$$TOL = \sum_{i=1}^{D} P_i - P_T - P_L. \tag{32}$$

Through comparison, we can find that the best, mean, worst costs and standard deviation obtained by MCSA are the least. Although the best, mean and worst costs are same for MABC and MCSA, the standard deviation, time and obtained by MCSA are much better than those by MABC. The standard CS and MCSA have almost the same effect on Case 1. Moreover, it shows that the MCSA is more consistent and stable than the other algorithms.

Fig. 3 shows the generation cost convergence of the best solution with iterations for CS ans MCSA for a typical run. It can be seen that both CS and MCSA enjoy smooth convergence, but MCSA is faster than CS.

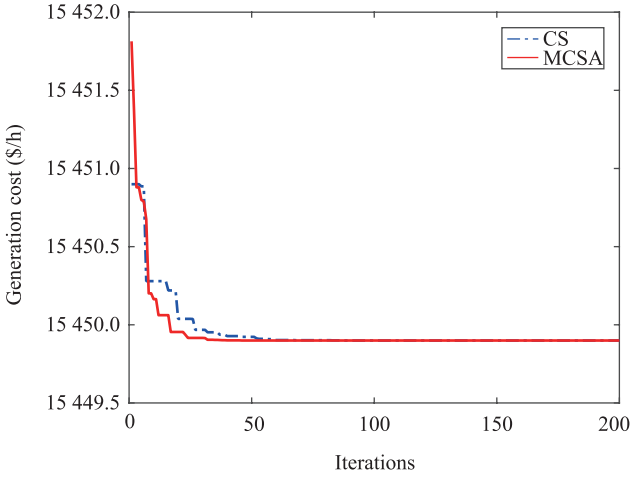


Fig. 3. Convergence characteristic of MCSA for Case 1.

C. System 2 (Cases 2 and 3: 13-unit)

System 2 includes 13 generators and supplies a total load demand of 2520 MW with valve-point effects but without ramp rate limits and prohibited operating zones. Moreover, this system is analyzed for both "without transmission losses (Case 2)" and "with transmission losses (Case 3)". The generator data of this system are taken from [64], and the loss coefficient is from [65] with correction $B_{i0}[11] = -0.0017$. This is a slightly larger system with more nonlinearities and local minima [66]. The best generation values and the costs of Cases 2 and 3 obtained by MCSA are presented in Table IV. Note that the generation values satisfy the generation limit constraints. The total generation costs of Cases 2 and 3 are 24 169.9177 $/h and 24 514.8756 $/h, respectively. The transmission loss of Case 3 is 40.4266 MW.

It can be seen from Table V that the statistical results obtained by MCSA are highly competitive compared to these by MABC [57], shuffled differential evolution (SDE) [65], one rank cuckoo search algorithm (ORCSA) [67], distributed sobol particle swarm optimization and tabu search algorithm (DSPSO-TSA) [68], cross entropy method and sequential quadratic programming (CE-SQP) [66], tabu search algorithm (TSA) [68], ant colony optimization (ACO) [69] and CS. MABC, SED, ORDSA and MCSA can get the best cost, but only MABC and MCSA get the optimal mean and the least

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TABLE III
COMPARISON BETWEEN MCSA AND OTHER PUBLISHED ALGORITHMS FOR CASE 1

No.	Algorithm	Best ($/h)	Mean ($/h)	Worst ($/h)	Std. dev.	Time (s)	*TOL* (MW)
1	**MCSA**	**15 449.8995**	**15 449.8995**	**15 449.8995**	**1.6404E−11**	**0.2599**	**−3.6380E−12**
2	MABC [57]	15 449.8995	15 449.8995	15 449.8995	6.04E−8	0.62	7.208E−11
3	BSA[58]	15 449.8995	15 449.9001	15 449.9056	0.0010	-	-
4	DHS [59]	15 449.8996	15 449.9264	15 449.9884	0.0204	0.01	8.89E−9
5	NPSO-LRS [60]	15 450	15 450.5	15 452	-	-	-
6	PSO[55]	15 450	15 454	15 492	-	14.89	-
7	MTS [61]	15 450.06	15 451.17	15 453.64	1.29	0.93	-
8	CBA [62]	15 450.2381	15 454.76	15 518.6588	2.965	0.704	-
9	CS [63]	15 449.8995	15 449.8995	15 449.8995	8.8315E−7	0.2514	−1.3642E−12

- Not reported in the referred literature.

TABLE IV
OPTIMAL GENERATIONS AND COST OBTAINED BY MCSA FOR CASES 2 AND 3

Unit	$P_i^{\min}$ (MW)	$P_i^{\max}$ (MW)	Generation (MW)	
			Case 2 (without TL)	Case 3 (with TL)
1	0	680	628.3185	628.3185
2	0	360	299.1993	299.1993
3	0	360	299.1993	299.1993
4	60	180	159.7331	159.7331
5	60	180	159.7331	159.7331
6	60	180	159.7331	159.7331
7	60	180	159.7331	159.7331
8	60	180	159.7331	159.7331
9	60	180	159.7331	159.7331
10	40	120	77.3999	77.3999
11	40	120	77.3999	113.1112
12	55	120	92.3999	92.3999
13	55	120	87.6845	92.3999
Cost ($/h)			24 169.9177	24 514.8756
Transmission loss (MW)				40.4266

cost. Although MABC has a good performance on standard deviation value, it usually takes much more time than MCSA.

Table VI shows a comparison among the results obtained by MCSA and other recently published stochastic methods such as oppositional real coded chemical reaction optimization (ORCCRO) [70], MABC [57], SDE [65], biogeography based optimization (BBO) [70], differential evolution with biogeography based optimization (DE-BBO) [70], improved coordinated aggregation with particle swarm optimization (ICA-PSO) [71], self-tuning hybrid differential evolution (STHDE) [16], hybrid differential evolution (HDE) [16], differential evolution (DE) [16] and CS. The best cost obtained from MCSA is the least in comparison with other methods except ORCCRO. It must be mentioned that the best cost value of ORCCRO with 24 513.91 ($/h) is obtained by adopting a higher tolerance ($|TOL_{\text{ORCCRO}}| = 0.98\,\text{MW} > |TOL_{\text{MCSA}}| = 4.5470\text{E}{-}11\,\text{MW}$) in [70]. Both MABC and MCSA can find the same best, mean and worst costs in Case 3, but MCSA is superior to MABC on the standard deviation and time.

Figs. 4 and 5 show the generation cost convergence of the best solution with iterations for CS and MCSA. It can be seen that both CS and MCSA enjoy smooth convergence, but MCSA converges to the optimal solution faster.

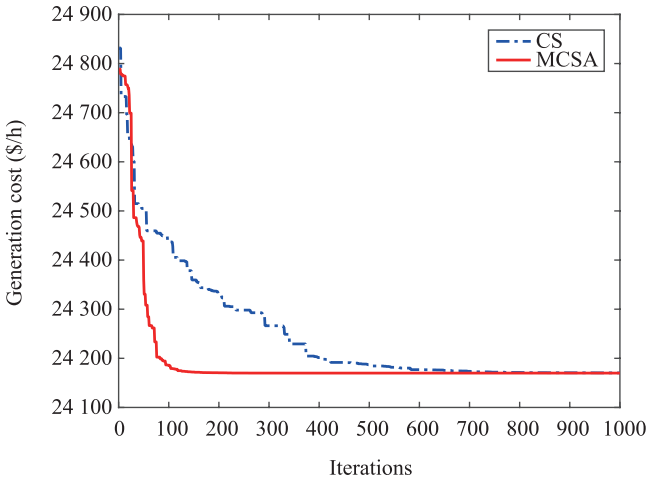


Fig. 4. Convergence characteristic of the MCSA for Case 2.

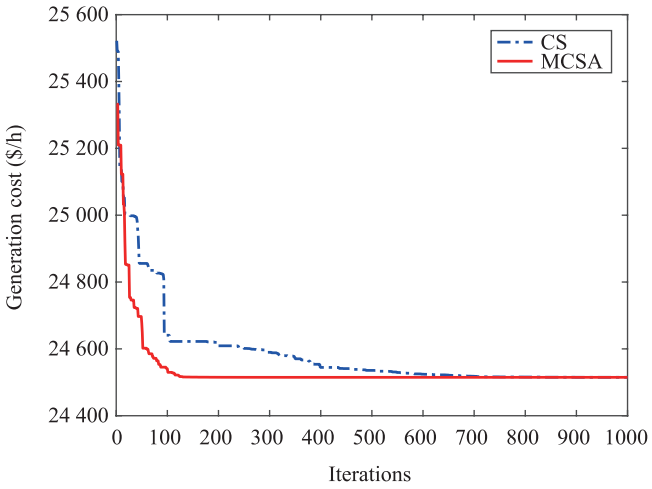


Fig. 5. Convergence characteristic of the MCSA for Case 3.

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TABLE V
COMPARISON AMONG MCSA AND OTHER PUBLISHED ALGORITHMS FOR CASE 2

No.	Algorithm	Best ($/h)	Mean ($/h)	Worst ($/h)	Std. dev.	Time (s)
1	**MCSA**	**24 169.9177**	**24 169.9177**	**24 169.9177**	**5.8558E−5**	**1.7152**
2	MABC [57]	24 169.9177	24 169.9177	24 169.9177	5.77E−7	32.7
3	SDE [65]	24 169.9177	24 170.7459	24 171.4402	-	-
4	ORCSA [67]	24 169.9177	24 182.2136	24 271.9232	21.988	3.22
5	DSPSO-TSA [68]	24 169.923	24 173.137	24 230.803	7.72	2.92
6	CE-SQP [66]	24 169.94	24 170.24	-	-	34.25
7	TSA [68]	24 171.211	24 184.055	24 392.203	41	10.52
8	ACO [69]	24 174.39	24 211.09	24 243.90	21.10	32.47
9	CS [63]	24 170.3175	24 173.6453	24 194.3547	3.5921	1.4767

TABLE VI
COMPARISON AMONG MCSA AND OTHER PUBLISHED ALGORITHMS FOR CASE 3

No.	Algorithm	Best ($/h)	Mean ($/h)	Worst ($/h)	Std. dev.	Time (s)	*TOL* (MW)
1	**MCSA**	**24 514.8756**	**24 514.8756**	**24 514.8756**	**3.1191E−7**	**2.5592**	**4.5470E−13**
2	ORCCRO [70]	24 513.91	24 513.91	24 513.91	-	-	0.98
3	MABC [57]	24 514.8756	24 514.8756	24 514.8756	3.5E−7	117.6*	2.91E−11
4	SDE [65]	24 514.88	24 516.31	-	-	-	-
5	BBO [70]	24 515.21	24 515.32	24 516.09	-	-	-
6	DE-BBO [70]	24 514.97	24 515.05	24 515.98	-	-	-
7	ICA-PSO [71]	24 540.06	24 541.46	24 589.45	-	-	-
8	STHDE [16]	24 560.08	24 706.63	24 872.44		2.9783	
9	HDE [16]	24 591.76	24 739.53	25 074.90		3.5733	
10	DE [16]	24 819.32	25 217.64	25 656.40		2.5815	
11	CS [63]	24 514.9857	24 516.9312	24 538.1519	3.9436	2.7166	0

This value is obtained by the authors and calculated through the results reported in [48].

D. System 3 (Case 4: 20-unit)

In this subsection, the ED problem is solved for a system with 20 generators considering the transmission losses and a demand of 2500 MW. VPE, RRL and POZ are neglected. The data of this test system can be found in [72]. Table VII shows the optimal generations and cost obtained by MCSA for Case 4. The optimal cost and corresponding transmission loss are 62 456.6331 $/h and 91.9667 MW, respectively.

Table VIII shows the comparison among MCSA and other methods such as CBA [62], continuous quick group search optimizer (CQGSO) [73], ORCSA [67], group search optimizer (GSO) [73], backtracking search algorithm (BSA) [74], biogeography based optimization (BBO') [75], Hopfield modeling (HM) [72], λ method [72] and CS. As observed from it, the best, mean and worst cost results of MCSA are better than those of all other methods including ORCSA, GSO, BSA, BBO', HM, and λ method and similar to those of CBA and CQGSO. The best, mean and worst costs of MCSA have the same values. Thus it verifies the robustness of the proposed method. Moreover, MCSA produces the optimal or the same standard deviation, mean and worst costs in comparison with those other than its peers. MCSA is more stable than the other optimization techniques regarding the mean cost and standard deviation. It also shows that MCSA and CS have same results for Case 4.

TABLE VII
OPTIMAL GENERATIONS AND COSTS OBTAINED BY MCSA FOR CASE 4

Unit	$P_i^{\min}$ (MW)	$P_i^{\max}$ (MW)	Generation (MW)
1	150	600	512.7817
2	50	200	169.1015
3	50	200	126.8907
4	50	200	102.8672
5	50	160	113.6829
6	20	100	73.5721
7	25	125	115.2900
8	50	150	116.3998
9	50	200	100.4049
10	30	150	106.0274
11	100	300	150.2385
12	150	500	292.7658
13	40	160	119.1142
14	20	130	30.8315
15	25	185	115.8059
16	20	80	36.2544
17	30	85	66.8592
18	30	120	87.9712
19	40	120	100.8027
20	30	100	54.3048
Cost ($/h)			62 456.6331
Transmission loss (MW)			91.9667

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TABLE VIII
COMPARISON AMONG MCSA AND OTHER PUBLISHED ALGORITHMS FOR CASE 4

No.	Algorithm	Best ($/h)	Mean ($/h)	Worst ($/h)	Std. dev.	Time (s)	*TOL* (MW)
1	**MCSA**	**62 456.6331**	**62 456.6331**	**62 456.6331**	**1.2077E−11**	**2.4668**	**−5.0022E−12**
2	CBA [62]	62 456.6328	62 456.6348	62 501.6714	0.3809	1.16	-
3	CQGSO [73]	62 456.6330	62 456.6331	62 456.6334	-	11.13	-
4	ORCSA [67]	62 456.6331	62 456.6331	62 456.6332	3E−5	0.32	-
5	GSO [73]	62 456.6332	62 456.6336	62 456.6353	-	30.45	-
6	BSA [74]	62 456.6925	62 457.1517	62 458.1272	-	14.562	-
7	BBO' [75]	62 456.7793	62 456.7928	62 456.7928	-	-	-
8	HM [72]	62 456.6341	-	-	-	-	-
9	λ method [72]	62 456.6391	-	-	-	-	-
10	CS [63]	62 456.6331	62 456.6331	62 456.6331	2.8818E−8	2.1913	−2.2732E−12

Fig. 6 shows the generation cost convergence of the best solution among the 50 trial runs of Case 4.

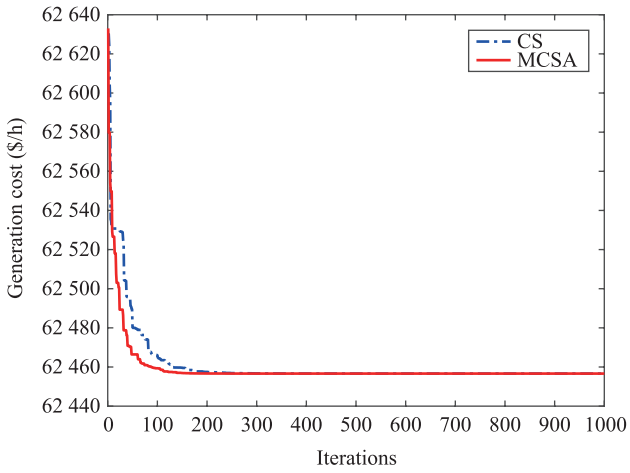


Fig. 6. Convergence characteristic of the MCSA for Case 4.

E. System 4 (Case 5: 40-unit)

In order to further demonstrate the efficiency and scalability of MCSA, a larger system with 40 units and VPE is considered. The demand of the system is 10 500 MW neglecting TL, RRL and POZ. The fuel costs and power generation limits are taken from [64]. Table IX presents the optimal generation values and cost obtained by MCSA. The optimal cost is 121 412.5355 $/h. It can be seen that the generations satisfy the generation limit constraints.

Table X shows the comparison of the results obtained by MCSA and other recently reported algorithms in the literature such as exchange market algorithm (EMA) [76], ORCSA [67], CSA [43], SDE [65], MABC [57], CBA [62], new adaptive particle swarm optimization (NAPSO) [77], species-based quantum particle swarm optimization (SQPSO) [78], CE-SQP [66], modified differential evolution (MDE) [79], firefly algorithm (FA) [80], BSA [74], QPSO [78], BBO' [75] and CS. It shows that the best cost result of MCSA is the same or better than those obtained with other methods. Also, the mean and worst costs obtained with MCSA are better than those from its peers. Moreover, its obtained standard deviation and time are better than those from most of other methods. In summary, MCSA is more consistent and stable than the other optimization techniques.

Convergence characteristic of MCSA for Case 5 is presented in Fig. 7. It shows that MCSA has better convergence characteristic in comparison with the standard CS.

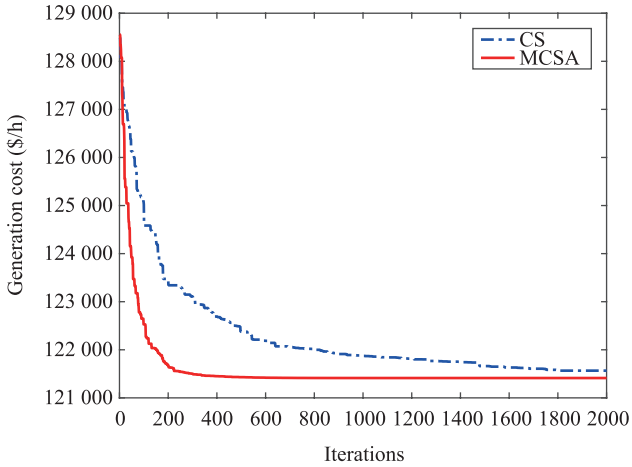


Fig. 7. Convergence characteristic of the MCSA for Case 5.

VI. CONCLUSION

In this paper, a modified CS algorithm is proposed. It is implemented to solve both convex and nonconvex economic dispatch problems by considering ramp rate limits, valve-point effects, transmission losses and prohibited operating zones. A slack method is used to handle equality constraints. A modified lambda iteration method is used to generate new solutions. Statistical results are compared with the reported results in literature. It is found that MCSA is capable of yielding a suitable balance between exploitation and exploration and has a better performance in terms of efficiency and robustness. All the experimental results confirm its high capability in solving ED problems. In the future, we need to develop more advanced efficient optimization methods [81]−[83] to solve power system problems involving renewable energy sources and parallel dispatch.

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TABLE IX
OPTIMAL GENERATIONS AND COST OBTAINED BY MCSA FOR CASE 5

Unit	$P_i^{\min}$ (MW)	$P_i^{\max}$ (MW)	Generation (MW)	Unit	$P_i^{\min}$ (MW)	$P_i^{\max}$ (MW)	Generation (MW)
1	36	114	110.7998	21	254	550	523.2794
2	36	114	110.7998	22	254	550	523.2794
3	60	120	97.3999	23	254	550	523.2794
4	80	190	179.7331	24	254	550	523.2794
5	47	97	87.7999	25	254	550	523.2794
6	68	140	140.0000	26	254	550	523.2794
7	110	300	259.5997	27	10	150	10.0000
8	135	300	284.5997	28	10	150	10.0000
9	135	300	284.5997	29	10	150	10.0000
10	130	300	130.0000	30	47	97	87.7999
11	94	375	94.0000	31	60	190	190.0000
12	94	375	94.0000	32	60	190	190.0000
13	125	500	214.7598	33	60	190	190.0000
14	125	500	394.2794	34	90	200	164.7998
15	125	500	394.2794	35	90	200	194.3978
16	125	500	394.2794	36	90	200	200.0000
17	220	500	489.2794	37	25	110	110.0000
18	220	500	489.2794	38	25	110	110.0000
19	242	550	511.2794	39	25	110	110.0000
20	242	550	511.2794	40	242	550	511.2794
		Cost ($/h)				121 412.5355	

TABLE X
RESULTS OF THE COMPARISON AMONG MCSA AND OTHER PUBLISHED ALGORITHMS FOR CASE 5

No.	Algorithm	Best ($/h)	Mean ($/h)	Worst ($/h)	Std. dev.	Time (s)
1	**MCSA**	**121 412.5355**	**121 414.1632**	**121 421.1219**	**2.7456**	**3.9948**
2	EMA [76]	121 412.5355	121 417.1328	121 426.1548	-	-
3	ORCSA [67]	121 412.5355	121 472.4534	121 596.1789	58.6005	3.02
4	CSA [43]	121 412.5355	121 520.4106	121 810.2538	81.5705	3.03
5	SDE [65]	121 412.54	121 415.72	121 418.58	-	-
6	MABC [57]	121 412.5409	121 431.7793	121 503.7552	19.16	115.2*
7	CBA [62]	121 412.5468	121 418.9826	121 436.15	1.611	1.55
8	NAPSO [77]	121 412.57	-	-	-	-
9	SQPSO [78]	121 412.5702	121 455.7003	121 709.5582	49.8076	47.24
10	CE-SQP [66]	121 412.88	121 423.65	-	-	-
11	MDE [79]	121 414.79	121 418.44	121 466.04	-	-
12	FA [80]	121 415.05	12 416.57	124 424.56	1.784	-
13	BSA [74]	121 415.6139	121 474.8823	121 524.9577	-	13.12
14	QPSO [78]	121 424.6399	121 586.9412	121 994.0267	114.080	48.25
15	BBO [75]	121 479.5029	121 512.0576	121 688.6634	-	-
16	CS [63]	121 566.0904	121 664.3187	121 794.7483	50.8508	4.0774

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